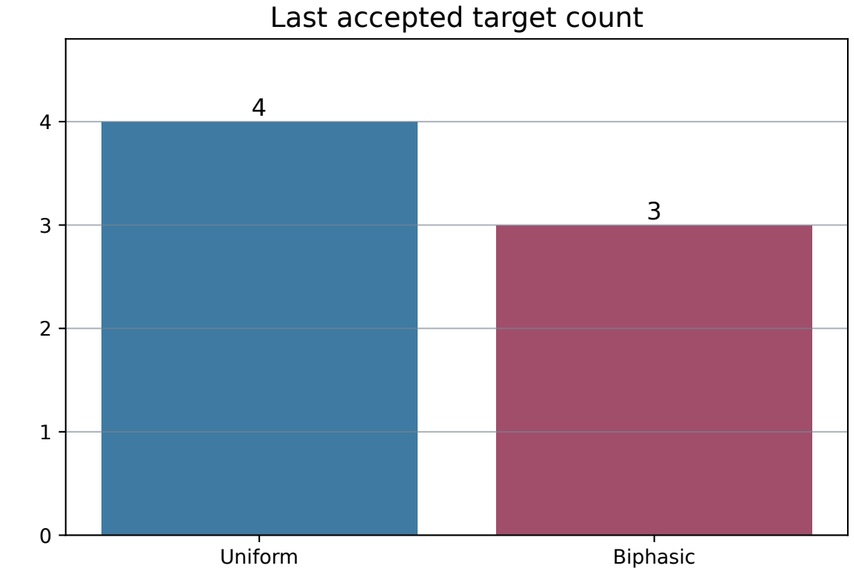
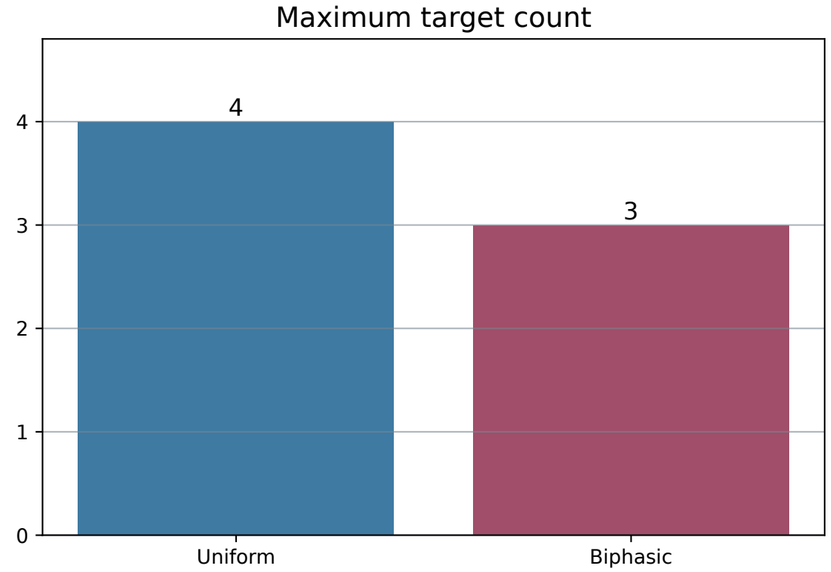
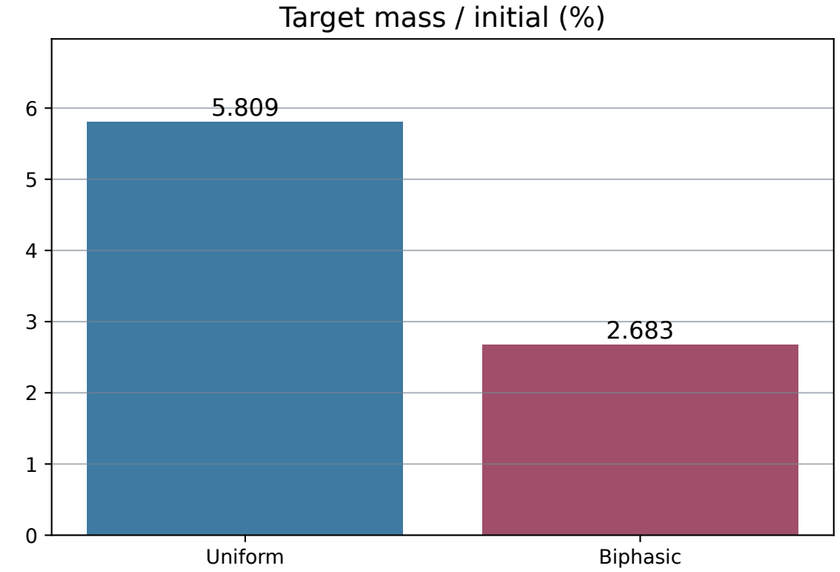
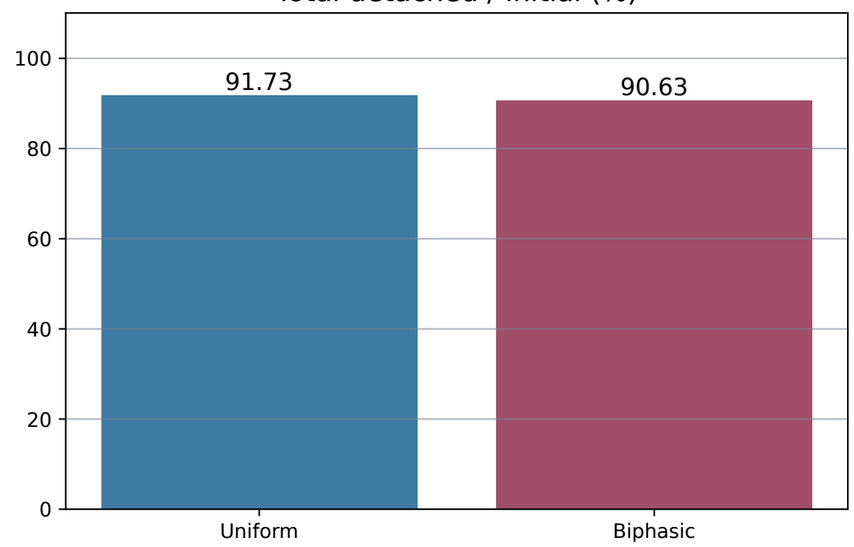
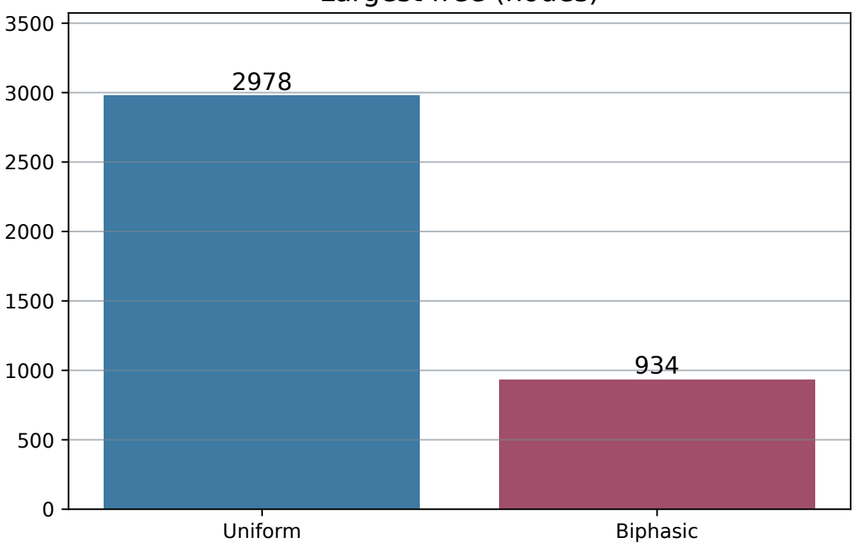
Scientific outcome from actual native trajectories · no grouping or prescribed fragment shapes
uniform: COMPLETED, 3.60000 ms | biphasic: COMPLETED, 3.60000 ms

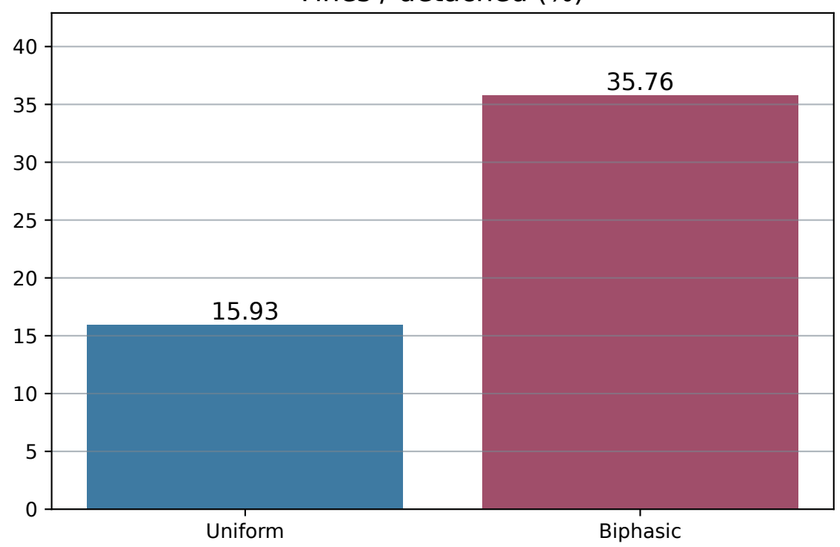


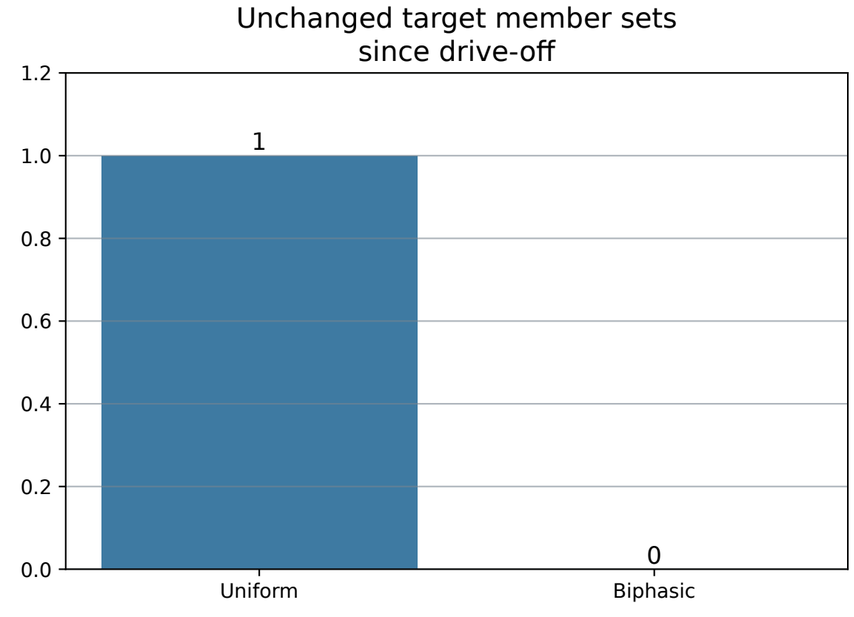


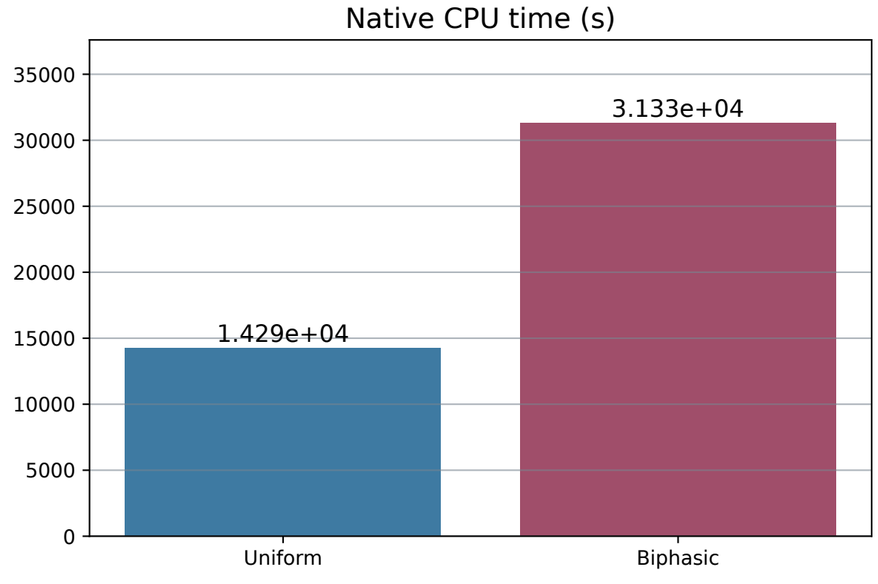












TARGET_NOT_ACHIEVED

Single frozen spatial realization
Uncalibrated material thresholds
Prescribed mechanical shear proxy
No validated bubble power
Graph separation ≠ certified
continuum crack surface